LECTURE 22

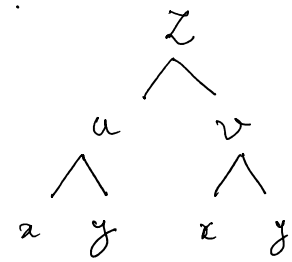
17 November 2023 09:42

$z = f(x, y)$ $\quad \frac{\partial z}{\partial x}$

Chain rule for partial derivatives : composite f^n.

$z = f(u, v)$ $\quad u = \phi(x, y)$ $\quad v = \psi(x, y)$

$\phi \to$ phi $\quad \psi \to$ psi



$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial x}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\frac{\partial v}{\partial y}$$

1) $z = \log(u^2 + v)$ $\quad u = e^{x+y^2}$ $\quad v = x^2 + y$ $\quad z = f(u, v)$

$\frac{\partial z}{\partial x}$ & $\frac{\partial z}{\partial y}$ → Find. $\quad \frac{\partial z}{\partial u}$

$$\frac{\partial z}{\partial u} = \frac{1}{u^2 + v} \cdot 2u \qquad \frac{\partial u}{\partial x} = e^{x+y^2} \qquad \frac{\partial z}{\partial v} = \frac{1}{u^2+v} \qquad \frac{\partial v}{\partial x} = 2x$$

$$\frac{\partial u}{\partial y} = e^{x+y^2} \cdot 2y \qquad \frac{\partial v}{\partial y} = 1$$

(*) $x^2 + y^2 - 1 = 0$ $\quad y = f(x)$ $\quad y = \pm\sqrt{1 - x^2}$ ✓ explicit f^n:

$x^2 + y^2 + 4xy + 25 = 0$ as $y = f(x)$

$y^5 + 2y^4 + 7y^3 + 6x^2 - 8x^3 + y = 0$?? as $y = f(x)$

f^n. which can't be expressed in $y = f(x)$ or $x = f(y)$ form are called implicit f^n:

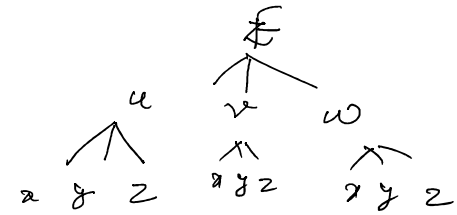
Given: $F(x, y) = 0$ Implicit f^n: $\quad \frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{\partial F/\partial x}{\partial F/\partial y}$

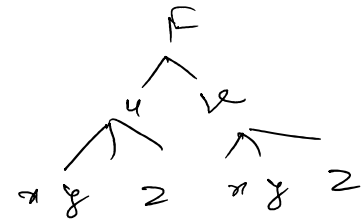
(1) $F(x, y) = e^y - e^x + xy$

$$\frac{\partial F}{\partial x} = 0 - e^x + y = y - e^x \qquad \frac{\partial F}{\partial y} = e^y - 0 + x \cdot 1 = e^y + x$$

$$\frac{dy}{dx} = -\frac{y - e^x}{e^y + x} = \frac{e^x - y}{e^y + x}$$

$\frac{\partial F}{\partial x} =$





<u>Directional derivative</u>: Let $P = (P_1, P_2)$ be any point. Then the D.D. of f at (a,b) in the direction of $P = (P_1, P_2)$ is

$$D_P f(a,b) = \lim_{s \to 0} \frac{f(a + sP_1,\ b + sP_2) - f(a,b)}{s}$$

$$\text{or} = \left(f_x P_1 + f_y P_2\right) \underset{\text{evaluate}}{\rightarrow} (a,b)$$

Polar form: D.D. of f at (a,b) in direction of θ

$$D_\theta f(a,b) = \lim_{t \to 0} \frac{f(a + t\cos\theta,\ b + t\sin\theta) - f(a,b)}{t}$$

$$\cos\theta = \frac{h}{\sqrt{h^2 + k^2}} \qquad \sin\theta = \frac{k}{\sqrt{h^2 + k^2}}$$

1) Find D.D. of $f(x,y) = x^2 + xy$ at $(1,2)$ in direction $\overset{P}{\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)}$

$$\left(\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}\right)$$

<u>Sol</u>: $f_x = 2x + y \qquad f_y = x$

$$D_P f(a,b) = (2x + y)\frac{1}{\sqrt{2}} + x \cdot \frac{1}{\sqrt{2}} = \frac{2}{\sqrt{2}}x + \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}$$

$$= \frac{3x}{\sqrt{2}} + \frac{y}{\sqrt{2}}$$

$$D_P f(1,2) = \frac{3}{\sqrt{2}} + \frac{2}{\sqrt{2}} = \frac{3}{\sqrt{2}} + \sqrt{2}$$

2) $f(x,y) = \dfrac{x^4+y^4}{x^2+y^2}$ $(x,y) \neq (0,0)$ ~~direction~~ $(0,0)$

0 $(x,y) = (0,0)$

$$D_\theta f(0,0) = \lim_{t\to 0} \frac{f(t\cos\theta,\ t\sin\theta)}{t} = \lim_{t\to 0} \frac{t^4(\cos^4\theta + \sin^4\theta)}{\cancel{t^2 \cdot t}}$$

$$= 0$$

3) $f(x,y) = \dfrac{x^4+y^4}{x^3+y^3}$

$$D_\theta f(0,0) = \lim_{t\to 0} \frac{\cancel{t^4}(\cos^4 + \sin^4\theta)}{\cancel{t^3}(\cos^3\theta + \sin^3\theta)}$$

In what directions D.D. will exist?? Find those θ.

Choose θ : $\cos^3\theta + \sin^3\theta \neq 0$

$\Rightarrow -\tan^3\theta \neq -1$

$\Rightarrow \tan\theta \neq -1$

$\Rightarrow \theta \neq \pi - \frac{\pi}{4}$ or $2\pi - \frac{\pi}{4}$